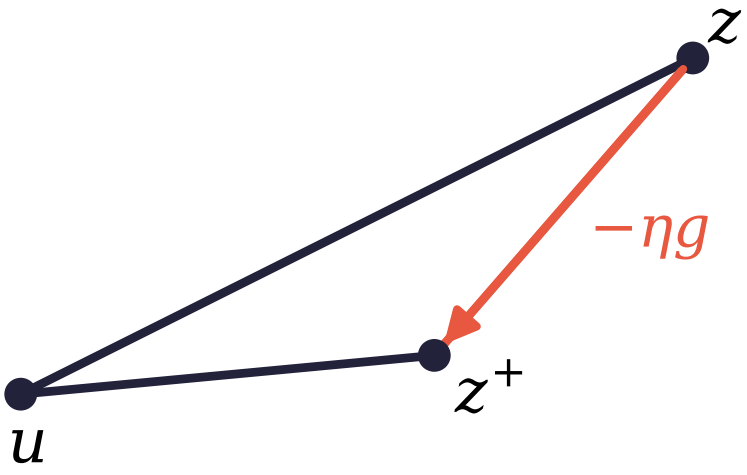
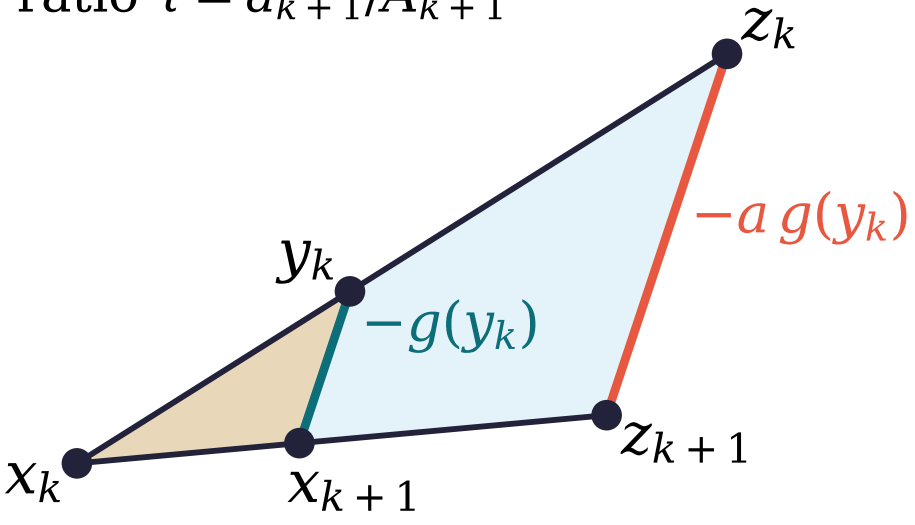
(a) three point identity

$2\langle z - z^+,\, u - z^+\rangle$

$= |u - z^+|^2 + |z - z^+|^2 - |u - z|^2$



(b) similar triangles

ratio $\tau = a_{k+1}/A_{k+1}$